

杭州电子科技大学学生考试卷（ A ）卷

考试课程	通信电路与系统	考试日期	年 月 日	成绩	
课程号	A040095s	教师号		任课教师姓名	李航, 孙朋飞
考生姓名		学号 (8 位)		年 级	专 业

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题
得分						

I . Single Choice (3' each)

- Which of the following circuits is not part of AM transmitting system? (B)

A. Local oscillator
B. Mixer

C. High frequency amplifier
D. Modulator
- When the operation frequency is lower than the resonance frequency, a series LC resonant circuit show (A).

A. Capacitive
B. Inductive

C. Resistor
D. All of the above
- To reduce electromagnetic radiation risk to human body, a maximum 23 dBm transmit power of 5G mobile phone is set, which corresponds to (B) mW unit.

A. 100
B. 200

C. 500
D. 1000
- When a transistor operates as a dependent-current source, what does the classification of power amplifier depend on (C)

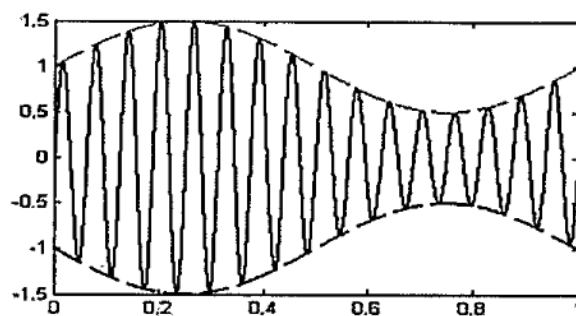
A. Output power
B. Magnification

C. Conduction angle
D. Amplitude of the input signal
- Quartz crystals working in the oscillator behave as (C)

A. Resistance
B. Open circuit

C. Inductor
D. Capacitor

6. As shown below, the modulation index of the AM signal, m_a , is (B)



A. $-1 < m_a < 0$

B. $0 < m_a < 1$

C. $m_a = 1$

D. $m_a > 1$

7. What is the startup condition of feedback oscillators? (n is an integer) (A)

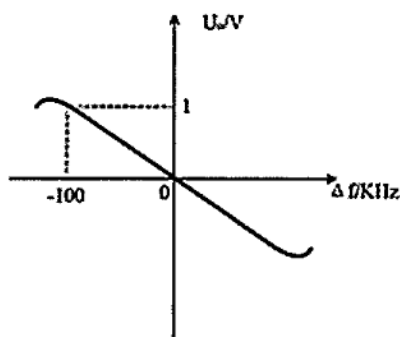
A. $A * F > 1, \phi_A + \phi_F = 2n\pi$

B. $A * F = 1, \phi_A + \phi_F = 2n\pi$

C. $A * F > 1, \phi_A + \phi_F = n\pi$

D. $A * F = 1, \phi_A + \phi_F = n\pi$

8. The frequency discrimination characteristics curve is shown below. If the output voltage of the frequency discriminator $u_o(t) = \cos 4\pi \times 10^4 t$ (V), the input signal $u_{FM}(t)$ can be expressed as (D)



A. $U_m \cos(2\pi f_c t + 5 \cos 4\pi \times 10^4 t)$ (V)

B. $U_m \cos(2\pi f_c t + 5 \sin 4\pi \times 10^4 t)$ (V)

C. $U_m \cos(2\pi f_c t - 5 \cos 4\pi \times 10^4 t)$ (V)

D. $U_m \cos(2\pi f_c t - 5 \sin 4\pi \times 10^4 t)$ (V)

9. Which of the following circuits is not part of a mixer? (D)

A. transistors

B. local oscillators

C. Intermediate frequency filter

D. Low frequency amplifier

10. By paralleling a resistor between two ends of LC resonant circuit, which of the following results is correct? (C)

A. Increase the quality factor

B. Increase the resonance frequency

C. Increase the bandwidth

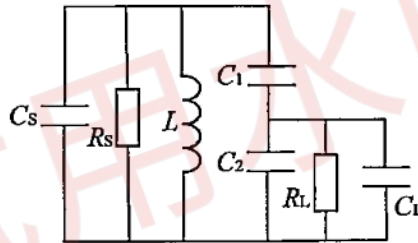
D. Decrease the bandwidth

II. Calculation

1. (8') Capacitor divider impedance conversion circuit is shown below, known that $L = 1 \mu\text{H}$, $Q_0 = 100$, $C_1 = C_2 = 10 \text{ pF}$, $C_s = 5 \text{ pF}$, $C_L = 24 \text{ pF}$, $R_s = 10 \text{ k}\Omega$, $R_L = 10 \text{ k}\Omega$.

(1) Calculate the resonance frequency f_0 and the resonant resistance R_0 (R_L and R_s are negligible).

(2) Calculate the loaded quality factor Q_L and bandwidth $B_{0.7}$.



$$R_L' = \frac{1}{n^2} R_L = \left(\frac{C_1 + C_2}{C_1} \right)^2 R_L = 40 \text{ k}\Omega$$

$$C_L' = n^2 \cdot C_L = 6 \text{ pF}$$

$$C_2 = C_s + C_L' + \frac{C_1 C_2}{C_1 + C_2} = 5 + 6 + 5 = 16 \text{ pF}$$

$$f_0 = \frac{1}{2\pi \sqrt{L C_2}} = 39.8 \text{ MHz}$$

$$R_0 = Q_0 \cdot 2\pi f_0 \cdot L = 25 \text{ k}\Omega$$

$$Q_L = \frac{Q_0}{1 + \frac{R_0}{R_s} + \frac{R_0}{R_L'}} = \frac{100}{1 + \frac{25\text{k}}{10\text{k}} + \frac{25\text{k}}{40\text{k}}} \approx 24.24$$

$$B_{0.7} = \frac{f_0}{Q_L} = \frac{39.8}{24.24} \approx 1.64 \text{ MHz}$$

2. (12') A resonant power amplifier with operating at critical point state. Assuming the output power $P_o = 16\text{W}$, $E_c = 20\text{V}$, current conduction angle $\theta = 70^\circ$, $\alpha_0(70^\circ) = 0.253$, $\alpha_1(70^\circ) = 0.436$, the parameters of the power amplifier: the slope of critical line $g_{cr} = 1.5\text{A/V}$ and $I_{cmax} < 5\text{A}$.

- (1) Please calculate the DC power P_{DC} , power dissipation at the collector P_c , collector efficiency η_c and optimal load resistor R_{cp} .
 (2) If the amplitude of input signal is doubled, how do the operating state and output power change?
 (3) To potentially prevent the transistor from burning out, what should we do for the equivalent load resistor?

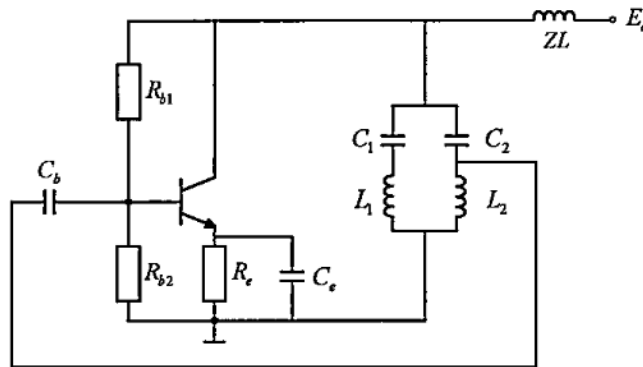
(1) $I_{cm} = \alpha_1 I_{cmax} = \alpha_1 g_{cr} (E_c - U_{cm})$ 1'
 $P_o = \frac{1}{2} I_{cm} U_{cm} = 0.327 \times (20 - U_{cm}) \cdot U_{cm} = 1.6$ 1'
 $\Rightarrow U_{cm} = 17.14\text{V}$ 1'
 $I_{cmax} = 1.5 \times (20 - 17.14) = 4.29 < 5\text{A}$ 1'
 $P_{DC} = E_c \cdot I_{c0} = 20 \times \alpha_0 \cdot I_{cmax} = 21.7\text{W}$ 1'
 $P_c = P_{DC} - P_o = 5.7\text{W}$ 1'
 $\eta_c = \frac{P_o}{P_{DC}} = 73.7\%$ 1'
 $R_{cp} = \frac{U_{cm}^2}{2P_o} = 9.18\Omega$ 1'

(2) The operating state will change to the overvoltage state.
 Because the current and voltage are almost unchanged,
 the output power is also almost unchanged. 2'

(3) We should prevent the equivalent load resistor from
 being too small. 2'

3. (12') The oscillation circuit is shown below, assuming $E_c = 10V$, $R_{b1} = 5 k\Omega$, $R_{b2} = 6 k\Omega$, $C_b = 0.2 \mu F$, $C_e = 0.1 \mu F$, $C_1 = 2000 pF$, $C_2 = 4000 pF$, $L_1 = 105 \mu H$, $L_2 = 15 \mu H$.

- (1) Draw the high frequency AC equivalent circuit, and explain the type of oscillator.
- (2) When drawing AC equivalent circuit, which components are equivalent to open circuit and which components are equivalent to short circuit?
- (3) Calculate the oscillating frequency f_0 and the feedback factor F .



(1) (Rb, Re are fine if added in the circuit)

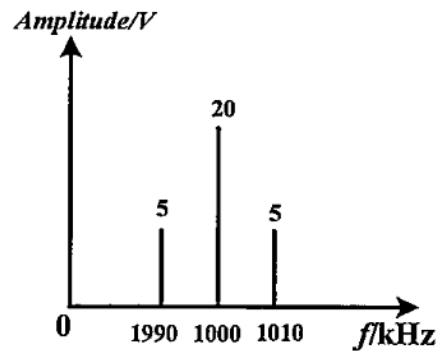
It is an inductance feedback oscillator (Hartley oscillator).

(2) Z_L is equivalent to open circuit.
 C_b and C_e are equivalent to short circuit.

(3) $L_{\Sigma} = L_1 + L_2 = 120 \mu H$
 $C_{\Sigma} = \frac{C_1 C_2}{C_1 + C_2} = 1333.3 pF$
 $f_0 = \frac{1}{2\pi \sqrt{L_{\Sigma} C_{\Sigma}}} = 397.9 kHz$
 $F = \frac{j2\pi f_0 L_2}{j2\pi f_0 L_1 + \frac{1}{j2\pi f_0 C_1}} = 0.6$

4. (8') The spectrum diagram of an AM signal is shown below

- (1) Please give the time-domain expression of this AM wave.
- (2) Please calculate the whole power with unit resistor, and the frequency bandwidth B.



(1) $U_{AM}(t) = 20 (1 + 0.5 \cos 2\pi \times 10^4 t) \cos 2\pi \times 10^6 t \text{ (V)}$

(2) $P_c = \frac{1}{2} \frac{U_{cm}^2}{R} = \frac{1}{2} \times 20^2 = 200 \text{ (W)}$

$P_{side} = \frac{1}{2} m_a^2 P_c = \frac{1}{2} \times 0.5^2 \times 200 = 25 \text{ (W)}$

$P_w = P_c + P_{side} = 225 \text{ (W)}$

$B = 2 \times 10 \times 10^3 = 20 \text{ kHz}$

5. (10') Known that the frequency and amplitude of a carrier signal are $f_c = 30$ MHz and $U_{cm} = 5$ V, respectively. The modulation signal is monofrequency, and its frequency $F = 500$ Hz, the maximum frequency shift $\Delta f = 15$ kHz.

- (1) Please give both expressions of FM and PM waves.
- (2) If F is changed to 3 kHz, and other parameters remain unchanged, please give both expressions of FM and PM waves.

(1) $U_{FM}(t) = 5 \cos(2\pi \times 30 \times 10^6 t + 30 \sin 2\pi \times 500 t)$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 U_{cm}, I' f_c, I' $\frac{\Delta f}{F}, I'$ F, I'

$U_{PM}(t) = 5 \cos(2\pi \times 30 \times 10^6 t + 30 \cos 2\pi \times 500 t)$

$\uparrow$ $\uparrow$
 I' I'

(2) $U_{FM}(t) = 5 \cos(2\pi \times 30 \times 10^6 t + 5 \sin 2\pi \times 3000 t)$

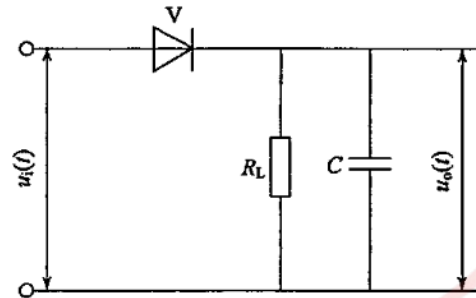
$\uparrow$ $\uparrow$
 I' I'

$U_{PM}(t) = 5 \cos(2\pi \times 30 \times 10^6 t + 30 \cos 2\pi \times 3000 t)$

$\uparrow$ $\uparrow$
 I' I'

6. (20') As shown in the figure below, assuming $R_L = 10 \text{ k}\Omega$, $m_a = 0.3$.

- (1) What kind of circuit is it? What function does it achieve?
- (2) If $f_c = 500 \text{ kHz}$ and the highest frequency of modulation signal $F = 300 \text{ Hz}$, how is the capacitor chosen to prevent the diagonal distortion?
- (3) If the detection efficiency $\eta_d = 0.95$, calculate the maximum equivalent input resistance R_{in} .
- (4) How is the input resistance of next-stage low-frequency amplifier chosen to prevent negative peak clipping distortion?
- (5) Can the circuit below be used to demodulate the DSB and SSB signals? If yes, please sketch its working principle. If not, what kind of other circuits can be selected commonly? and draw its circuit.



1) It is a large signal peak envelope detector.
It achieves the function of demodulating (or detecting) the standard AM signal.

$$(2) R_L C \gg T = \frac{1}{f_c} \Rightarrow C \gg \frac{1}{f_c R_L} = \frac{1}{500 \text{ kHz} \cdot 10 \text{ k}\Omega} = 200 \text{ pF}$$

$$C < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{F \cdot R_L} = \frac{\sqrt{1-0.3^2}}{0.3} \times \frac{1}{300 \times 10^3 \times 10^4} = 0.169 \mu\text{F}$$

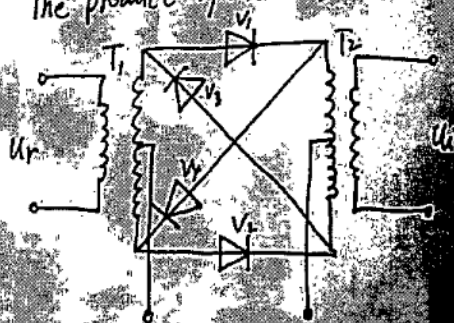
$$200 \text{ pF} \ll C < 0.169 \mu\text{F}$$

$$(3) R_{in} \approx \frac{R_L}{2\eta_d} = \frac{10 \text{ k}\Omega}{2 \times 0.95} = 5263.2 \Omega$$

$$(4) m_a \leq \frac{R_i}{R_i + R_L} \Rightarrow R_i \geq \frac{m_a R_L}{1 - m_a} = \frac{0.3 \times 10 \text{ k}\Omega}{1 - 0.3} = 4285.7 \Omega$$

(5) It can not demodulate DSB and SSB signals.

The product synchronous demodulation circ. can be selected.



杭州电子科技大学学生考试卷（ B ）卷

考试课程	通信电路与系统	考试日期	年 月 日		成绩	
课程号	A040095s	教师号		任课教师姓名	李航, 孙朋飞	
考生姓名		学号 (8 位)		年级		专业

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题
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I. Judgment questions (3' each)

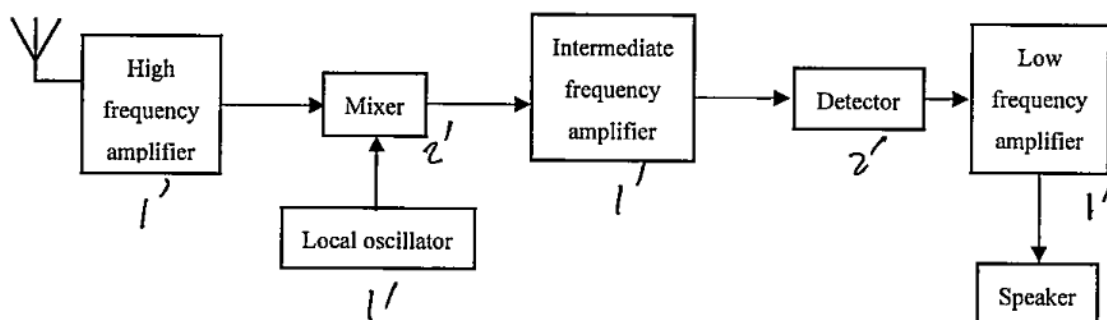
1. Television, FM radio and mobile communications belong to shortwave (HF) communications. (×)
2. When the operating frequency is smaller than the resonant frequency, the LC resonant circuit is only inductive. (×)
3. For Class C power amplifier, the conduction angle is less than π . (√)
4. Paralleling a resistor between two ends of LC resonant circuit will increase the quality factor. (×)
5. Hartley circuit has better output waveform than Colpitts circuit. (×)
6. Overtone crystal oscillator has higher oscillation frequency than Quartz crystal oscillator. (√)
7. Angle modulation circuit is a linear spectrum shifting circuit. (×)
8. SSB receiver has higher SNR than AM receiver, because it has only half bandwidth of AM receiver. (√)
9. The frequency offset in an FM signal is related to the frequency of modulation signal. (×)
10. Compared with AM, FM signals are power saving and bandwidth wasting. (√)

II. Calculation

1. (10') Please draw the compositions diagram of superheterodyne receiving system.

Please choose a frequency band for IF of AM radio in China in parentheses.

(465 kHz, 10.7 MHz or 38 MHz)

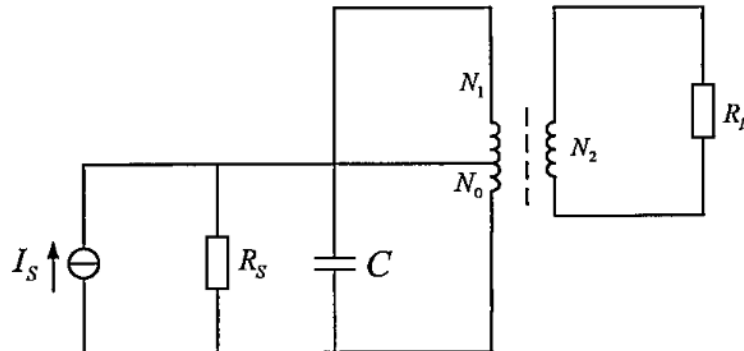


465 kHz

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2. (8') As shown below, known the resonant circuit $f_0 = 455 \text{ kHz}$, $Q_0 = 100$, the access factor in primary coils $n_1 = N_1/N_0 = 4$, the access factor between primary and secondary coils $n_2 = N_1/N_2 = 16$, $C = 200 \text{ pF}$, $R_S = 16 \text{ k}\Omega$ and $R_L = 1 \text{ k}\Omega$.

- (1) Calculate the inductor L and the resonant resistance R_0 (R_L and R_S are negligible).
- (2) Calculate the loaded quality factor Q_L and bandwidth $B_{0.7}$.



$$\begin{aligned}
 (1) \quad L &= \frac{1}{(2\pi f_0)^2 C} = \frac{1}{(2\pi \times 455 \text{ kHz})^2 \cdot 200 \times 10^{-12}} = 612.39 \text{ }\mu\text{H} \\
 R_0 &= \frac{Q_0}{2\pi f_0 C} = 174.98 \text{ k}\Omega \\
 R_\Sigma &= R_0 \parallel R'_S \parallel R'_L \\
 R'_S &= n_1^2 R_S = 256 \text{ k}\Omega \\
 R'_L &= n_2^2 R_L = 256 \text{ k}\Omega \\
 R_\Sigma &= 73.92 \text{ k}\Omega, \quad Q_L = \frac{R_\Sigma}{\omega_0 L} = 42.2 \\
 B &= \frac{f_0}{Q_L} = \frac{455}{42.2} = 10.78 \text{ kHz}
 \end{aligned}$$

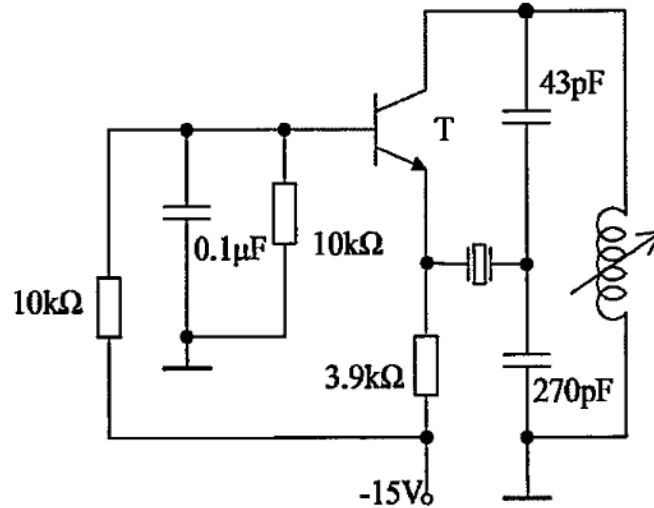
3. (10') A resonant power amplifier with operating at critical state. Assuming $E_c = 24$ V, current conduction angle $\theta = 70^\circ$, $\alpha_0(70^\circ) = 0.253$, $\alpha_1(70^\circ) = 0.436$, $I_{cmax} = 2.2$ A and the slope of critical line $g_{cr} = 0.8$ A/V.

- (1) Please calculate the DC power P_{DC} , power dissipation at the collector P_c , collector efficiency η_c and optimal load resistor R_{cp} .
 (2) If the input power $P_{in} = 30$ dBm, calculate the power gain in dB.

$$\begin{aligned}
 (1) \quad P_{DC} &= E_c \cdot I_{c0} = E_c \cdot I_{cmax} \alpha_0(70^\circ) = 24 \times 0.253 \times 2.2 = 13.36 \text{ W} & 1' \\
 U_{cem} &= E_c - U_{ces} = E_c - \frac{I_{cmax}}{g_{cr}} = 24 - \frac{2.2}{0.8} = 21.25 \text{ V} & 2' \\
 P_o &= \frac{1}{2} U_{cem} \cdot I_{cm} = \frac{1}{2} U_{cem} \cdot I_{cmax} \alpha_1(70^\circ) = 10.19 \text{ (W)} & 1' \\
 P_c &= P_{DC} - P_o = 3.17 \text{ W} & 1' \\
 \eta_c &= \frac{P_o}{P_{DC}} = 0.76 & 1' \\
 R_{cp} &= \frac{U_{cem}^2}{2P_o} = 22.16 \Omega & 1' \\
 (2) \quad P_o &= 10.19 \text{ W} \rightarrow P_o = 40.1 \text{ dBm} & 1' \\
 G_{dB} &= 40.1 - 30 = 10.1 \text{ dB} & 2'
 \end{aligned}$$

4. (10') The oscillation circuit is shown below, assuming the crystal oscillation frequency $f = 25 \text{ MHz}$, $L = 1 \mu\text{H}$.

- (1) Draw its AC equivalent circuit, explain the role of crystal and the type of oscillator.
- (2) Calculate the oscillating frequency f_0 and the feedback factor F of the circuit.



(d)

The crystal is equivalent to a short circuit.

It is a quartz crystal oscillator.

(2) $f_0 = 25 \text{ MHz}$

$$F = \frac{U_f}{U_o} = \frac{\frac{1}{j2\pi f \cdot 270\text{pF}}}{\frac{1}{j2\pi f \cdot 43\text{pF}} + \frac{1}{j2\pi f \cdot 270\text{pF}}} = 0.14$$

4'

1'

1'

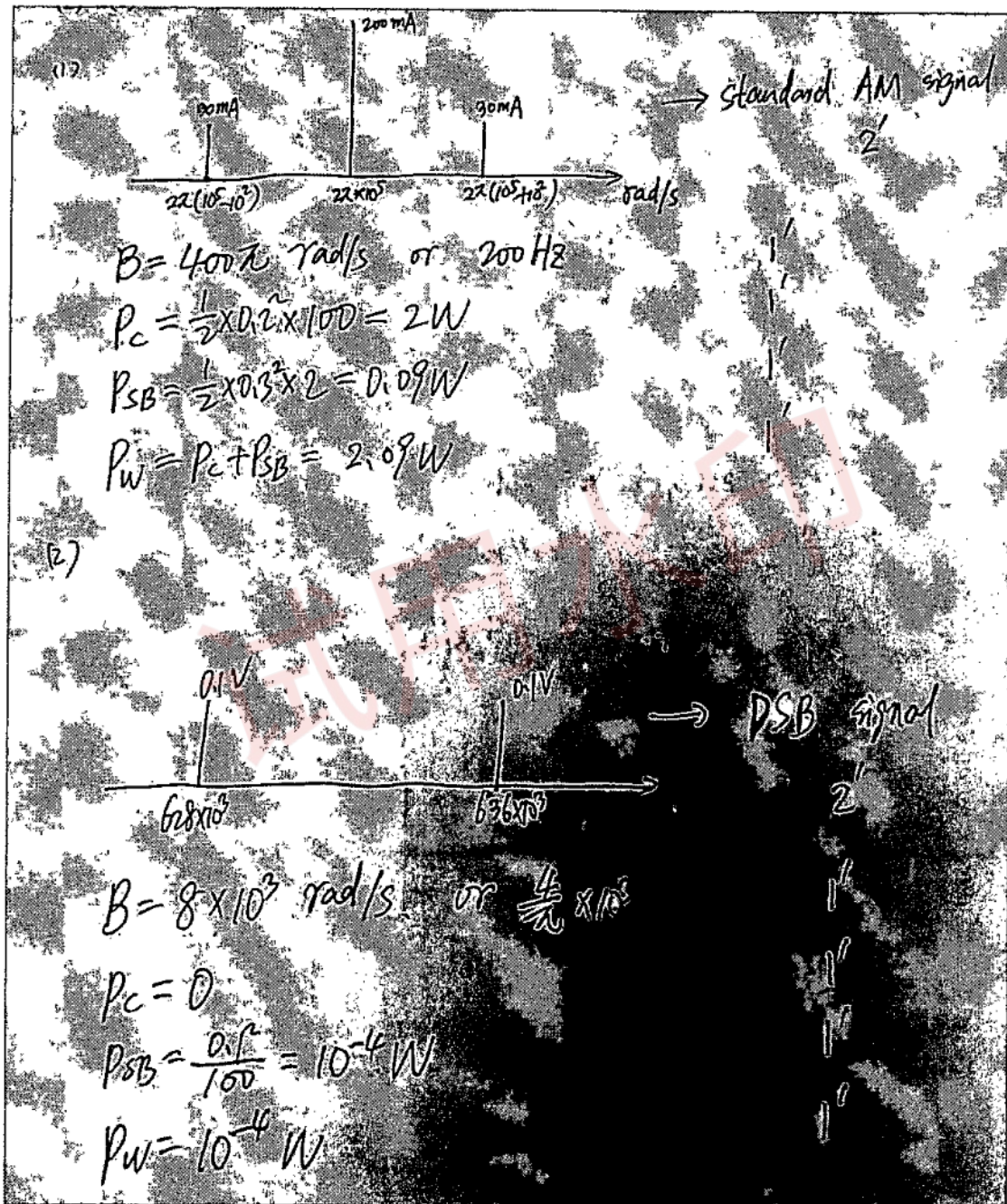
2'

2'

5. (12') Please draw the spectra of the following two AM signals, and explain what kind of AM signals are they? Calculate the bandwidth B in rad/s, the carrier power P_c , the side frequency power P_{SB} and the whole power P_w , where the load resistance $R_L = 100\Omega$.

(1) $i(t) = 200(1 + 0.3\cos 200\pi t)\cos 20000\pi t$ mA.

(2) $u(t) = 0.1\cos 628 \times 10^3 t + 0.1\cos 636 \times 10^3 t$ V



6. (20) Known that the frequency of modulation signal is 400 Hz, its amplitude is 2.4 V and the angle modulation index is 60. When the frequency of modulation signal is reduced to 250 Hz and its amplitude is increased to 3.2

- (1) For FM signals, calculate its angle modulation index and bandwidth.
- (2) For PM signals, calculate its angle modulation index and bandwidth;
- (3) For FM and PM signals, when the modulation amplitude is fixed, sketch the maximum phase shift ($\Delta\phi$) and the maximum frequency shift ($\Delta\omega$) vs. the frequency of modulation signal (Ω), respectively;
- (4) Please draw a FM equivalent AC circuit to realize the direct FM.
- (5) To demodulate FM signals, please draw the circuit schematic of slope frequency discriminator.

